

DS-First Outline

Ian Palmer

Tentative Outline

Framing: This project is aiming to be a practical guide to a newly-accessible federal dataset, built around real exploratory findings. The current domain problem is that it is difficult for energy economists to study how firms contract for electricity, because the main dataset recording it has been generally unusable. Now that the data has been made more accessible, there are still barriers to use that an economist might run into. The guide is how I address that.

1. Introduction: A Dataset Nobody Can Use

Sets up the domain problem and explains why the data is interesting.

Almost all wholesale electricity in the US is sold through privately negotiated contracts rather than on public markets, so the prices and terms are hard to observe. FERC has collected every one of these contracts since 2013, but published them in a format so unwieldy that only a handful of papers have ever used the data.

States what the project delivers: a guide, a set of worked analyses, and a tool.

2. Background: How Wholesale Electricity Gets Bought and Sold

Short, non-technical orientation for readers without an energy background, including me. Covers what wholesale means, the difference between regions with organized markets and regions without, and why bilateral contracts exist. Defines the terminology needed to read the rest of the paper.

3. The Data: FERC's Electric Quarterly Reports

Describes the dataset itself. Who is required to file and who is exempt (notably all of Texas), what the four tables contain, what a row means in each one, and which variables carry the most information. Explains the relationship between contracts and transactions, and cites both FERC and PUDL.

4. Getting the Data: Working with 100 GB from a Laptop

The computational methods section. Explains the Parquet format and why columnar storage makes a dataset this size queryable without downloading it, then walks through querying it remotely with SQL through DuckDB in R. Includes the filter-aggregate-then-pull-down pattern that makes everything else possible.

Show with examples with annotated code.

5. What's Actually in There: Data Quality and Validation

Addresses the data concerns that were posted on the PUBL website.

Documents what I found when checked the data rather than trusting it at face value, walk through my barriers working with the data: contracts are restated every quarter and must be de-duplicated, several variables show discontinuities around 2018–19 that look like reporting changes rather than real behavior, and some price fields contain values that can't be literal. Gives future users a list of which variables are trustworthy and under what conditions.

*Note these are just a few possibilities that I have noticed looking at the data. I need to spend more time working with the data to see if they are worth exploring and what else is important to note.

6. Worked Examples: Questions This Data Can Answer

Four or five short analyses, possibly more. Each posing an energy-economics question and showing the code and result. Current options: how contract length has changed over a decade, whether market-based and cost-based sellers commit differently, whether contracts with corporate affiliates differ from arm's-length ones, and how prices vary by region. The last example should apply the method to a current important question to show how the dataset can be used on live policy questions (possibly around data centers or current US conflicts).

7. An Interactive Tool

*Note: this is a idea, I need to talk this through with my advisor and professor more to see what is plausible/would be useful.

A dashboard that lets someone explore contract trends by region, date range, and contract type without writing any code. Built for the non-technical domain expert, the economist who knows what's interesting but doesn't want to learn SQL.

8. Ethical Considerations

Identify issues specific to this project. Some ideas I have already are the filer identity table contains named individuals' work emails, phone numbers, and addresses, which is lawfully public but not obviously appropriate to republish. Another is that there's a question about making commercially sensitive contract data easier to analyze, whether easier access to competitors' pricing serves transparency or could facilitate coordination.

9. Limitations and Future Work

What the dataset can't do: no plant identifiers or locations, no link to other energy datasets, no Texas, and no retail sales. Notes that FERC Order 917 changes the filing regime in 2026 and removes several fields, which makes this guide a record of a closing data regime. Identifies the some next project steps: like building a crosswalk between EQR companies and EIA plant data.

10. References